

LLM-Based Inductive Synthesis of a Local Behavior Tree for Decentralized Multi-Robot Cooperation



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Introduction

Motivation

- Robots sharing the same goal differ in position, role, and observations, so per-robot control logic is often hand-designed, making it hard to scale as the number of robots or the environment changes.
- Behavior Trees (BTs)** help modularize robot behavior, but deploying a different BT per robot raises management cost and limits reusability.

Objectives

- Deploy the **Identical BT** to every robot using only local observations, achieving decentralized cooperative multi-robot control without per-robot BT design.
- Use an LLM to inductively synthesize the Identical BT from per-robot demonstrations, improving reusability across diverse tasks and environments.

Methods

Global Plan

- A* computes, for each cube, a path from its start to its goal, serving as the reference for Contribution Check.

Identical BT (3 child nodes)

- Contribution Check** — Checks whether the robot's own actions can contribute to the given global plan.
- Assign Check** — When neighboring robots independently assign themselves to the same task, a decentralized Multi-Robot Task Allocation (MRTA) algorithm resolves the conflict.
- Problem Solving BT** — The LLM-synthesized local action subtree that executes the assigned task.

Two-Stage Inductive Synthesis

- Stage 1:** Guided by a global plan, the LLM generates a subtree for each robot.
- Stage 2:** These robot-specific subtrees are lifted to variables and inductively generalized into a single Problem Solving BT usable by every robot.

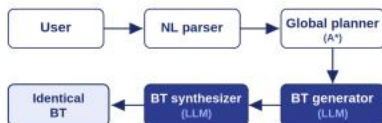


Figure 1. From user command to Identical BT

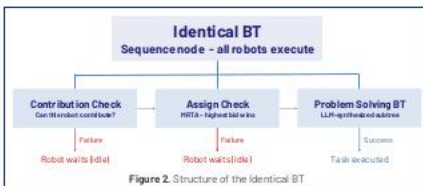


Figure 2. Structure of the Identical BT

Experiment

Setup

- MuJoCo simulator; ROBOTIS OMX-F robots fixed on an $n \times n$ grid; each does Pick & Place only within its adjacent 2×2 cells; ROS2 Action Server drives the BT in a **Cube Relay** scenario.
- Baseline **One-Step**: Single LLM call generates one BT XML per condition (9 conditions: grid $4 \times 4 / 5 \times 5 / 6 \times 6 \times$ cubes $2/3/4$).

Method	2 Cubes	3 Cubes	4 Cubes	Total
One-Step	3/3	2/3	0/3	5/8
Ours	3/3	3/3	3/3	9/9

Table 1. Success by number of cubes (One-Step vs. Ours)

- One BT, synthesized from the $4 \times 4 / 2$ -cube condition with a single synthesis call, solved **all 9 conditions (9/9)** with no extra LLM calls; whereas One-Step requires a separate generation per condition and degrades as cubes grow.
- Beyond these 9 conditions, **same BT scaled to a 7×7 grid with 16 cubes** without modification, showing reusability and scalability.



Figure 3. Visualization in the MuJoCo-based simulator

Future Work

- Replace the path-planning A* with a general AI task planner to generate global plans for arbitrary task domains.
- Validate the framework through diverse experiments to demonstrate that it generalizes across a wide range of environments.
- Deploy the framework on real OMX robots to verify its effectiveness beyond simulation.